

Derbyshire Shared Care Pathology Guidelines

Interpretation of prolactin results

Introduction

Prolactin is synthesised and secreted by the lactotroph cells of the anterior pituitary gland under the inhibitory control of the hypothalamus via dopamine release. Prolactin release can be induced by oestrogen, thyrotropin releasing hormone and dopamine antagonism.

The main physiological function of prolactin is the induction and maintenance of lactation, hence the expected physiological rise in prolactin levels seen during pregnancy and breastfeeding. Prolactin release varies during circadian rhythm and is found to be most elevated overnight/ during sleep. This can lead to a physiological elevation in prolactin in early morning blood samples. Physiological rise in prolactin levels is also seen in times of stress including following difficult venepuncture attempts.

Pathological causes of hyperprolactinaemia include hypothalamic causes, pituitary causes, pharmacological causes and systemic causes.

Potential symptoms associated with hyperprolactinaemia include galactorrhoea, oligomenorrhoea/amenorrhoea, reduced libido/erectile dysfunction, hirsutism and infertility. Hyperprolactinaemia in the context of a pituitary macroadenoma may also present with headaches, visual changes and symptoms of other pituitary hormone deficiencies.

Causes of hyperprolactinaemia

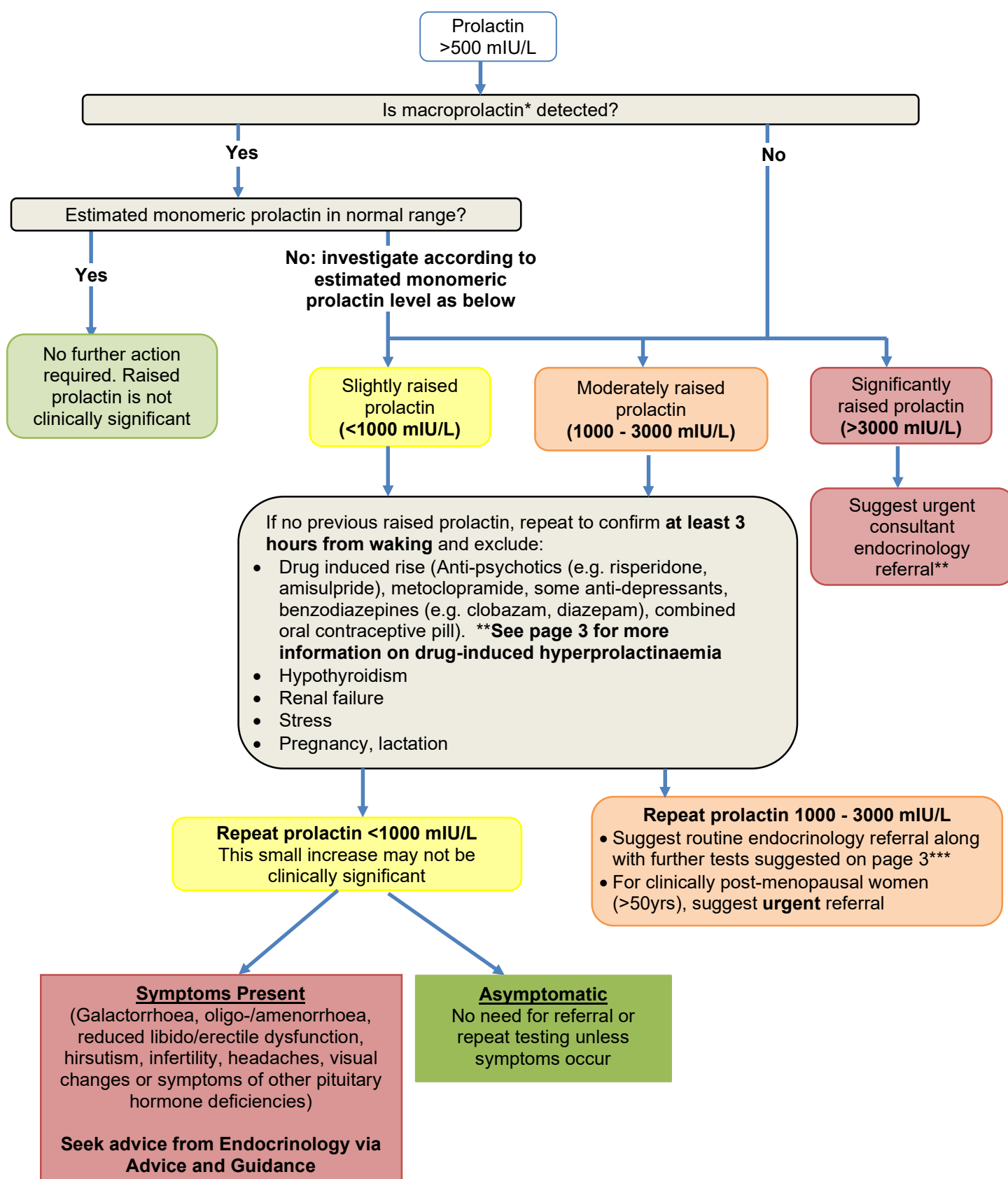
Physiological	Pharmacological	Pathological
• Stress • Pregnancy • Lactation • Seizures • Sleep	• Antipsychotics (especially risperidone, amisulpride, haloperidol) • Antidepressants • Peripheral dopamine receptor blockers (e.g. metoclopramide) • Histamine Antagonists (H2) • Antihypertensives (e.g. calcium channel antagonists) • Oestrogens (e.g. COCP) • Opiates	• Prolactinoma • Other pituitary or hypothalamus disease • Hypothyroidism • Renal failure • Liver failure • PCOS

Prolactin Reference Ranges

Males	86 – 324	mIU/L
Females	102 – 496	mIU/L

Note that prolactin levels below the reference range are generally not significant and do not require investigation.

Investigation and referral for raised prolactin



*Macroprolactin

'Macroprolactin' is a circulating complex of prolactin bound to IgG, reported to have a prevalence of up to 5.8% in the general population, but which is thought to have little bioactive activity and therefore be of little pathological significance.

Macroprolactin is known to cross-react in the prolactin assay and, if present, can significantly contribute to the measured prolactin result producing raised serum values. It is therefore necessary to check for the presence of macroprolactin so that patients are not over investigated for this benign condition.

The lab will check for macroprolactin in all patients when measured prolactin is >500 mIU/L. If macroprolactin is detected the lab will report an estimate of the normal, bioactive monomeric prolactin level. Once excluded, investigation for macroprolactin will not be repeated in subsequent samples.

**Anti-psychotic induced hyperprolactinaemia further points

- Antipsychotic induced hyperprolactinaemia is usually less than 3000 mIU/L, but drugs like risperidone and amisulpride can be associated with much higher levels
- Baseline prolactin levels pre-initiation of antipsychotic are strongly recommended to help identify medication induced hyperprolactinaemia. **NICE CG178 (Psychosis and schizophrenia in adults) does not recommend routine monitoring of prolactin levels in asymptomatic patients.** If the patient becomes symptomatic, repeat prolactin levels alongside other relevant investigations.
- Anti-psychotic agents shouldn't be stopped without discussion with Psychiatry – an alternative agent with less dopamine agonist activity should be considered where feasible (e.g. aripiprazole, quetiapine). Where able to stop potentially interfering medication, suggest repeat prolactin levels 1 week post-cessation (or 3 months post depot injection).
- Consider referral or advice request to endocrinology if prolactin is >3000 mIU/L or if prolactin <3000 mIU/L associated with symptoms suggestive of pressure effects such as headache, visual disturbances or amenorrhoea.

***Additional investigations recommended in patients referred to Endocrinology

- Early morning Cortisol (expedite referral if <250 nmol/L)
- TSH, FT3/FT4 (expedite referral if FT4 <10 mIU/L)
- LH, FSH and Testosterone/Oestradiol

Contacts

Derby

Duty Biochemist	01332 789383 (8am to 7pm, Mon – Fri)
On Call Consultant Biochemist	Via RDH switchboard, 01332 340131
Endocrinology Advice (urgent)	07879 115507 (9am – 5pm, Mon – Fri)
Endocrinology advice (non-urgent)	Choose 'Advice and Guidance' from NHS e-Referrals

Chesterfield

Duty Biochemist	01246 512212 (9am to 5pm, Mon – Fri)
On Call Consultant Biochemist	Via CRH switchboard, 01246 277271
Endocrinology Advice (urgent)	01246 513104
Endocrinology advice (non-urgent)	Choose 'Advice and Guidance' from NHS e-Referrals

Burton

Duty Biochemist	01283 593313 (8.30 am -4pm Mon-Fri), diverts to RDH duty biochemist out of these hours
On call Consultant Biochemist	Via QHB Switchboard 01283 566333
Endocrinology Advice (urgent)	Via QHB Switchboard
Endocrinology advice (non-urgent)	Choose 'Advice and Guidance' from NHS e-Referrals

References:

1. Diagnosis and Treatment of Hyperprolactinemia: An Endocrine Society Clinical Practice Guideline, Melmed *et al.* 2011 *The Journal of Clinical Endocrinology & Metabolism*, **96.2** 273–288
2. Prolactin Secretion and Sleep, Spiegel K *et al.* 1994 *Sleep* **17** 20–27.
3. Prolactin Biology and Laboratory Measurement: An update on Physiology and Current Analytical Issues, Saleem M *et al.* 2018 *Clinical Biochemist Reviews* **39** 3–16.
4. Peveler RC, Branford D, Citrome L *et al.* (2008). Antipsychotics and hyperprolactinaemia: clinical recommendations. *J Psychopharmacology* 22(2 Suppl):98-103.
5. NICE CG178 Psychosis and schizophrenia in adults: prevention and management (March 2014)

Document control

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Secondary Care UHDB	16.06.26	Dr R Stanworth	Consultant Endocrinologist
Secondary Care CRH	Feb 23	Dr R Robinson	Consultant Endocrinologist
Primary Care	18.07.25	Dr P Blackwell	GP
Laboratory (Derbyshire Pathology)	16.06.26	Ms S Knowles	Consultant Clinical Scientist